Changmin **Lee**

(+82) 10-9288-8981 | lchm1106@unist.ac.kr | changmin-lee.com | CkdalsKong | changmin0lee | Changmin Lee | UAI Lab, UNIST

Education

Ulsan National Institute of Science and Technology (UNIST) *Ulsan, Korea*

Integrated M.S./Ph.D. Program, Computer Science and Engineering *Mar. 2025 – Present*

- UAI Lab, Advisor: Prof. Taesik Gong
- Research focus: Personalized AI, on-device RAG, efficient LLMs, and AI agents.

Kyungpook National University *Daegu, Korea*

B.S., Computer Science and Engineering *Mar. 2019 – Feb. 2025*

- Total GPA: **3.83/4.3** Major GPA: **4.07/4.3**

Research Interests

- **Personalized AI & RAG** (user-specific memory construction, preference-aligned retrieval)
- **On-Device AI** (efficient LLM inference under tight resource budgets)
- **AI Agents & Memory** (compact memory for GUI agents, long-horizon task planning)

Research Experience

UAI Lab, UNIST *Ulsan, Korea*

Graduate Researcher *Mar. 2025 – Present*

- Conduct research on personalized AI, on-device RAG, efficient memory systems, and resource-efficient AI agents.
- Developed EPIC, a preference-aligned memory construction and retrieval framework for personalized RAG under on-device resource constraints. EPIC improved preference-following accuracy by 18.79 percentage points on average across four benchmarks while achieving 2,404× lower memory usage and 32.17× lower retrieval latency than the strongest baseline. The work was accepted at **ICML 2026** and formed the basis of a Korean patent application.
- Investigate transition-guided memory and flexible action reuse for efficient mobile GUI agents, aiming to reduce redundant model inference during task execution.

Pattern Recognition & Machine Intelligence Lab, Kyungpook National University *Daegu, Korea*

Undergraduate Research Assistant *Dec. 2023 – Sep. 2024*

- Conducted research on explainable AI, computer vision, medical image analysis, and machine learning.
- Developed a hybrid Transformer-CNN model for chest X-ray classification.
- Participated in regular research paper seminars and presented recent papers in computer vision and medical AI.

Publications

- **"From Volume to Value: Preference-Aligned Memory Construction for On-Device RAG,"**
  Changmin Lee, Jaemin Kim, and Taesik Gong,
  *International Conference on Machine Learning (**ICML**)*, 2026.
- **"Enhancing Chest X-Ray Classification with Multi-Class Token Hybrid Transformers,"**
  Changmin Lee, Ho-kyung Shin, and Woo-Jeoung Nam,
  Journal of The Institute of Electronics and Information Engineers, 2024.
- **"Design and Optimization of a Face Recognition-Based Entry Logging System to Prevent Solitary Deaths,"**
  Changmin Lee, Hyun-Jin Dong, Ju-Yong Park, Jin-Seong Bae, *et al.*,
  Korean Institute of Information Technology Fall Conference, 2024.
- **"Comparison of Recycling Waste Awareness Performance by Transfer Learning Model,"**
  Dong-hyuk Kim, Changmin Lee, *et al.*,
  Korean Multimedia Society Conference, 2024.

Manuscripts in Preparation

- **"ΔAgent: Efficient Mobile GUI Agents via Transition-Guided Flexible Action Reuse,"**
  Changmin Lee, Taehwan Park, Jongwon Lee, and Taesik Gong,
  2026. *Manuscript in preparation*.

Patent Applications

- **"Preference-Based Index Management Method for Generating Personalized Search Responses and System Supporting the Same,"**
  Taesik Gong, Changmin Lee, and Jaemin Kim,
  Republic of Korea Patent Application No. 10-2025-0182629, filed Nov. 26, 2025. *Patent pending*.

Awards & Fellowships

Aug. 2025 – Sep. 2026	**NRF Master's Fellowship**, Graduate Student Research Encouragement Grant, National Research Foundation of Korea
2024	**Excellence Award**, Undergraduate Research Support Program, Convergent Medical Scientist Training Program
Nov. 2024	**Outstanding Paper Award**, Undergraduate Paper Competition, Korean Institute of Information Technology Fall Conference

Graduate Project Experience

Dependency-Aware Compensation Scheduling for Multi-Step AI Agent Workflows

Team Member — *Fall 2025*

- Co-developed DACS, a user-space scheduler that compensates for dependency-induced and I/O-induced waiting in heterogeneous AI agent workflows.
- Designed and evaluated QoS-aware ticket scheduling across CPU, GPU, and I/O resources on an NVIDIA Jetson Orin Nano.

Personalized Mixture-of-Experts Model for User-Specific Adaptation

Team Member — *Jun. – Sep. 2025*

- Developed a mixture-of-experts model for personalized adaptation as part of the LG Electronics HS Future Technology Project.
- Designed the adaptation strategy, experimental setup, and evaluation pipeline.

Human Action Recognition with YOLO-Pose and CNN-GCN Fusion

Team Leader — *Spring 2025*

- Led the development of an end-to-end pipeline for person detection, pose estimation, and four-class action recognition.
- Proposed and implemented a CNN-GCN fusion model combining MobileNetV2 visual features with graph-based pose features, achieving a CodaLab score of 0.89.

Undergraduate Project Experience

Raspberry Pi-Based Face Recognition Entry Logging System

Team Member — *Sep. – Dec. 2024*

- Developed a real-time face recognition and entry logging system using Raspberry Pi, DeepFace, Flask, and a database-backed web service for solitary death prevention.
- Evaluated multiple face recognition models and similarity metrics, identifying FaceNet512 with Euclidean L2 as the best-performing combination with 94.8% accuracy.

Real-Time Recycling Waste Detection with Transfer Learning

Team Member — *Mar. – Jun. 2024*

- Constructed and refined a recycling waste dataset by establishing labeling criteria and expanding the labeled data from approximately 2,000 to 6,000 images.
- Fine-tuned and evaluated YOLOv9, YOLOv8, and RT-DETR for real-time waste detection, with YOLOv9 selected as the best-performing model.

Smart Table Clock

Team Member — *Sep. – Dec. 2023*

- Arduino-based smart clock with LCD widgets, Daegu bus API integration, and Wi-Fi web control.

Remote File Explorer

Team Leader — *Mar. – Jun. 2023*

- Built a terminal-based remote file explorer in C using the curses library with Linux-style commands.
- Enabled file browsing and transfer between macOS and Linux systems over a network.

Teaching Experience

Artificial Intelligence Fundamentals, AI Novatus Academia (Gyeongnam), 7th Cohort

Teaching Assistant — *May – Jun. 2026*

- Supported lectures and student exercises on fundamental AI concepts.

On-Device AI for Smart Manufacturing (NOVA508), UNIST

Teaching Assistant — *Spring 2026*

- Supported project-based activities and hands-on practice in on-device AI deployment.
- Assisted students with assignments and implementation-related questions.

Data Preprocessing and Visualization, AI Novatus Academia & LG Electronics LDC

Teaching Assistant — *2025 – 2026*

- AI Novatus Academia (Gyeongnam): 6th Cohort (May 2025), 7th Cohort (May–Jun. 2026)
- AI Novatus Academia (Ulsan): 8th Cohort (Jul. 2025)
- LG Electronics LDC Program: Jul. 2025 and Jan. 2026
- Assisted with data cleaning, transformation, exploratory data analysis, and visualization.

Computer Networks (CSE351), UNIST

Teaching Assistant — *Fall 2025*

- Supported course operations, grading, and student communication.
- Assisted students with course materials, assignments, and project questions.

Industry-Academic Project Mentoring

Strawberry Smart Farm Growth Cycle Prediction, AI Novatus Academia (Ulsan), 8th Cohort

Project Mentor — *Sep. – Dec. 2025*

- Mentored a team developing a growth-cycle classification model for smart-farm strawberry cultivation.

Vacuum Cleaner Personalization Pipeline, LG Electronics DIC Program PBL

Project Mentor — *Aug. – Sep. 2025*

- Mentored a team designing an end-to-end personalization pipeline for LG vacuum cleaner user scenarios.

Cutting Tool Dynamic Characteristics Prediction, AI Novatus Academia (Gyeongnam), 6th Cohort

Project Mentor — *Jun. – Sep. 2025*

- Mentored a team developing a model to predict dynamic characteristics of machining tools from cutting-process images.

Leadership & Service

UAI Lab, UNIST — *Ulsan, Korea*

Student Representative — *Jan. – Dec. 2025*

- Coordinated lab operations, internal communication, meeting schedules, and administrative tasks.
- Supported onboarding and communication among lab members.

Skills

Programming	Python, C, Java
Machine Learning	PyTorch, TensorFlow, Keras, scikit-learn
RAG & Retrieval	FAISS, dense retrieval, LLM-based evaluation, personalized RAG
Systems & Tools	Linux, Git, Docker, LaTeX, vLLM